

Cover Page

Standard: BACB30VF Rev: (Y) 04-Nov-2013

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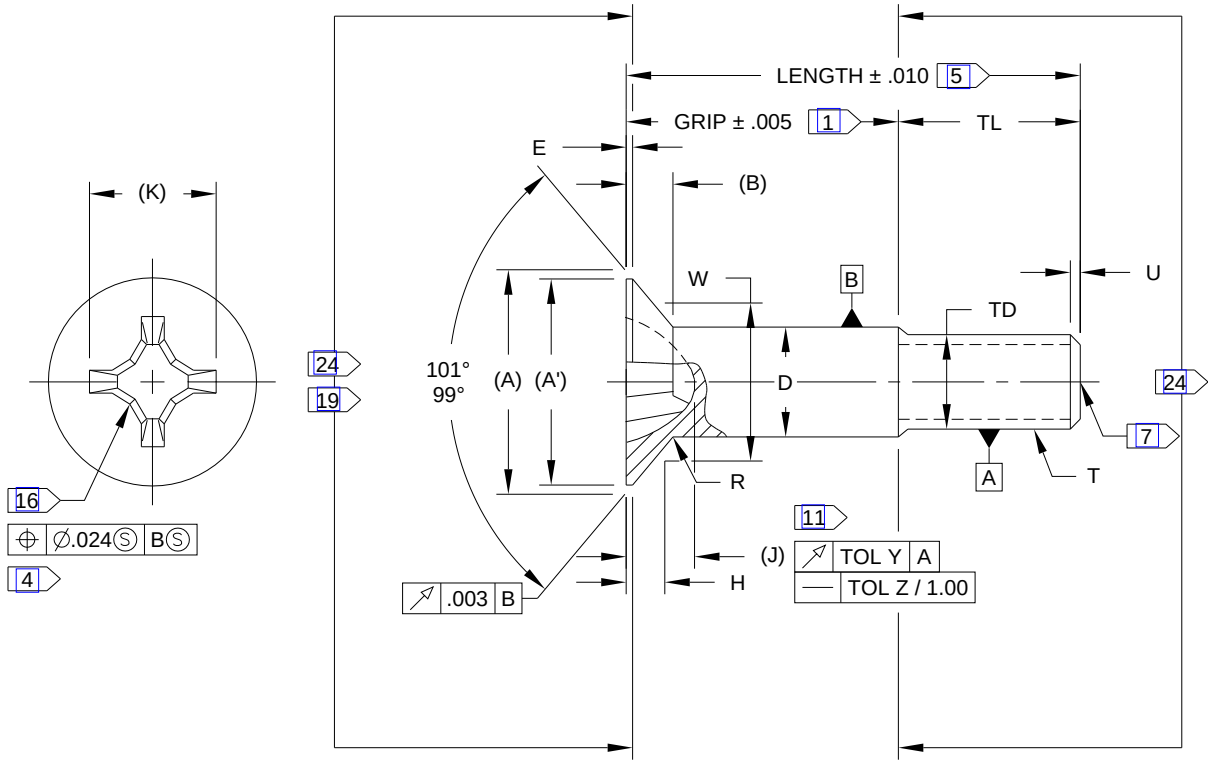
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FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

BCA	P	BDS	P								
NEW DESIGN APPROVAL: P=PARTIAL, F=FULL, N=NONE											



DIMENSIONING AND TOLERANCING PER [ASME Y14.5M-1994](#).
DIMENSIONS IN INCHES UNLESS OTHERWISE NOTED.
DIMENSIONS APPLY AFTER FINISH UNLESS OTHERWISE SPECIFIED.

TABLE I NOMINAL BOLT DIMENSIONS

BOEING STANDARD NUMBER BACB30VF 8	T NOM THREAD SIZE 6	Ø A TO SHARP CORNER MAX REF 10	Ø A' MIN REF 10	B MAX REF 10	Ø D			
					BEFORE FINISH	AFTER FINISH		
						ALL	EXCEPT PHOSPHAE FLUORIDE	PHOSPHATE FLUORIDE
					MIN	MAX	MIN	MIN
08	.1640-32	.263	.231	.042	.1621	.1635	.1625	.1630
3	.1900-32	.303	.271	.048	.1881	.1895	.1885	.1890
4	.2500-28	.397	.364	.063	.2481	.2495	.2485	.2490

TECHNICAL CHANGES IDENTIFIED BY REVISION BAR.

DATE 02-JUN-1982 REV (Y) 04-NOV-2013

CAGE CODE 81205

BACB30VF
SH 1 OF 12

BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)

BACB30VF
SH 1 OF 12

BOEING PART STANDARD

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TABLE I NOMINAL BOLT DIMENSIONS (CONTINUED)

BOEING STANDARD NUMBER BACB30VF [8]	T NOM THREAD SIZE [6]	Ø A TO SHARP CORNER MAX REF [10]	Ø A' MIN REF [10]	B MAX REF [10]	Ø D			
					BEFORE FINISH	AFTER FINISH		
						ALL	EXCEPT PHOSPHAE FLUORIDE	PHOSPHATE FLUORIDE
					AL COAT			
					MIN	MAX	MIN	MIN
5	.3125-24	.477	.442	.070	.3106	.3120	.3110	.3115
6	.3750-24	.564	.529	.080	.3731	.3745	.3735	.3740
7	.4375-20	.673	.629	.100	.4356	.4370	.4360	.4365
8	.5000-20	.755	.704	.108	.4981	.4995	.4985	.4990

TABLE I NOMINAL BOLT DIMENSIONS (CONTINUED)

BOEING STANDARD NUMBER BACB30VF 8	Ø D NO FINISH		E		H GAGE PROTRUSION 3		J MAX REF	Ø K MAX REF	R RAD	
	MAX	MIN	MAX	MIN	NOM	± TOL			MAX	MIN
08	.1635	.1630			.0201	.0014	.086	.154		
3	.1895	.1890			.0230	.0015	.101	.169	.020	.010
4	.2495	.2490	.010	.003	.0255	.0017	.116	.184		
5	.3120	.3115			.0280	.0018	.135	.247	.025	
6	.3745	.3740			.0305	.0019	.168	.317		
7	.4370	.4365	.012	.005	.0400	.0025	.193	.342	.030	.015
8	.4995	.4990	.015		.0408	.0025	.218	.366		

TABLE I NOMINAL BOLT DIMENSIONS (CONTINUED)

BOEING STANDARD NUMBER BACB30VF 8	TL NOM 5	TD 6		U MAX	Ø W GAGE +.0002 -.0000 3	TOL Y	TOL Z	
		MAX	MIN					
08	.312	.1595	.1570	.031	.2109	.0045	.0040	
3	.323	.1840	.1810		.2438			
4	.370	.2440	.2410	.036	.3312			.0030
5	.438	.3060	.3020	.042	.4045			
6	.454	.3680	.3640		.050	.4852	.0060	.0025
7	.528	.4310	.4260	.5696				

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CAGE CODE 81205

BACB30VF

SH 2

BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)

BACB30VF

SH 2

BOEING PART STANDARD

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TABLE I NOMINAL BOLT DIMENSIONS (CONTINUED)

BOEING STANDARD NUMBER BACB30VF [8]	TL NOM [5]	TD [6]		U MAX	Ø W GAGE +.0002 -.0000 [3]	TOL Y	TOL Z
		MAX	MIN				
8	.528	.4930	.4880	.050	.6498	.0060	.0020

TABLE I NOMINAL BOLT DIMENSIONS (CONTINUED)

BOEING STANDARD NUMBER BACB30VF [8]	RECESS NUMBER PER NAS7100 [16]	RECESS TORQUE IN-LB MIN	TENSILE STRENGTH LBS MIN
08	2	25	1,160
3		37	1,800
4		50	3,300
5	3	125	4,500
6	4	230	6,480
7		300	9,000
8		440	12,150

TABLE II OVERSIZE BOLT DIMENSIONS [15]

BOEING STANDARD NUMBER BACB30VF [8]	T NOM THREAD SIZE [6]	Ø A TO SHARP CORNER MAX REF [10]			Ø A' MIN REF [10]		B MAX REF [10]			
		.0156 OVERSIZE		.0312 OVERSIZE	.0156 OVERSIZE		.0312 OVERSIZE	.0156 OVERSIZE		.0312 OVERSIZE
		STD HEAD [12]	B CODE [13]		STD HEAD [12]	B CODE [13]		STD HEAD [12]	B CODE [13]	
08 [9]	.1640-32	---	---	---	---	---	---	---	---	---
3	.1900-32	.303	.319	.335	.271	.279	.301	.043	.049	.049
4	.2500-28	.397	.413	.429	.364	.372	.394	.056	.063	.062
5	.3125-24	.477	.493	.509	.442	.450	.471	.063	.070	.070
6	.3750-24	.564	.580	.596	.529	.537	.559	.074	.080	.080
7	.4375-20	.673	.689	.705	.629	.637	.660	.093	.100	.099

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BACB30VF

SH 3

BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)

BACB30VF

SH 3

BOEING PART STANDARD

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SEE APPLICABILITY BLOCK

TABLE II OVERSIZE BOLT DIMENSIONS 15 (CONTINUED)

BOEING STANDARD NUMBER BACB30VF 8	T NOM THREAD SIZE 6	Ø A TO SHARP CORNER MAX REF 10			Ø A' MIN REF 10			B MAX REF 10		
		.0156 OVERSIZE		.0312 OVERSIZE	.0156 OVERSIZE		.0312 OVERSIZE	.0156 OVERSIZE		.0312 OVERSIZE
		STD HEAD 12	B CODE 13		STD HEAD 12	B CODE 13		STD HEAD 12	B CODE 13	
		8	.5000-20	.755	.771	.787	.704	.712	.735	.101

TABLE II OVERSIZE BOLT DIMENSIONS 15 (CONTINUED)

BOEING STANDARD NUMBER BACB30VF 8	Ø D							
	.0156 OVERSIZE				.0312 OVERSIZE			
	BEFORE FINISH	AFTER FINISH			BEFORE FINISH	AFTER FINISH		
	AL COAT	ALL	EXCEPT PHOSPHATE FLUORIDE	PHOSPHATE FLUORIDE	AL COAT	ALL	EXCEPT PHOSPHATE FLUORIDE	PHOSPHATE FLUORIDE
	MIN	MAX	MIN	MIN	MIN	MAX	MIN	MIN
08 9	---	---	---	---	---	---	---	---
3	.2012	.2026	.2016	.2021	.2168	.2182	.2172	.2177
4	.2637	.2651	.2641	.2646	.2793	.2807	.2797	.2802
5	.3262	.3276	.3266	.3271	.3418	.3432	.3422	.3427
6	.3887	.3901	.3891	.3896	.4043	.4057	.4047	.4052
7	.4512	.4526	.4516	.4521	.4668	.4682	.4672	.4677
8	.5137	.5151	.5141	.5146	.5293	.5307	.5297	.5302

TABLE II OVERSIZE BOLT DIMENSIONS 15 (CONTINUED)

BOEING STANDARD NUMBER BACB30VF 8	Ø D			
	.0156 OVERSIZE (CODE "X")		.0312 OVERSIZE (CODE "Y")	
	NO FINISH		NO FINISH	
	MAX	MIN	MAX	MIN
08 9	---	---	---	---
3	.2026	.2021	.2182	.2177
4	.2651	.2646	.2807	.2802

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BACB30VF

SH 4

BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)

BACB30VF

SH 4

BOEING PART STANDARD

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FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

TABLE II OVERSIZE BOLT DIMENSIONS 15 (CONTINUED)

BOEING STANDARD NUMBER BACB30VF 8	Ø D			
	.0156 OVERSIZE (CODE "X")		.0312 OVERSIZE (CODE "Y")	
	NO FINISH		NO FINISH	
	MAX	MIN	MAX	MIN
5	.3276	.3271	.3432	.3427
6	.3901	.3896	.4057	.4052
7	.4526	.4521	.4682	.4677
8	.5151	.5146	.5307	.5302

TABLE II OVERSIZE BOLT DIMENSIONS 15 (CONTINUED)

BOEING STANDARD NUMBER BACB30VF 8	E		H GAGE PROTRUSION 3					J MAX REF	Ø K MAX REF	R RAD	
			.0156 OVERSIZE			.0312 OVERSIZE					
	MAX	MIN	NOM		TOL ±	NOM	TOL ±			MAX	MIN
			STD HEAD 12	B CODE 13							
08 9	---	---	---	---	---	---	---	---	---	---	---
3	.010	.003	.0230	.0296	.0015	.0361	.0015	.101	.169	.020	.010
4			.0255	.0321	.0017	.0386	.0017	.116	.184		
5			.0280	.0346	.0018	.0411	.0018	.135	.247	.025	
6	.012	.005	.0305	.0371	.0019	.0436	.0019	.168	.317	.030	.015
7			.0400	.0466	.0025	.0531	.0025	.193	.342		
8			.0408	.0474	.0025	.0539	.0025	.218	.366		

TABLE II OVERSIZE BOLT DIMENSIONS 15 (CONTINUED)

BOEING STANDARD NUMBER BACB30VF 8	TL (NOM) 5	TD 6		U MAX	Ø W GAGE +.0002 -.0000 3	TOL Y	TOL Z	RECESS NUMBER PER NAS7100 16	RECESS TORQUE IN-LB MIN	TENSILE STRENGTH LBS MIN
		MAX	MIN							
08 9	---	---	---	---	---	---	---	---	---	---
3	.323	.1840	.1810	.031	.2438	.0045	.0040	2	37	1,800
4	.370	.2440	.2410	.036	.3312		.0030		50	3,300
5	.438	.3060	.3020	.042	.4045			3	125	4,500

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BACB30VF

SH 5

BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)

BACB30VF

SH 5

BOEING PART STANDARD

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SEE APPLICABILITY BLOCK

TABLE II OVERSIZE BOLT DIMENSIONS ¹⁵ (CONTINUED)

BOEING STANDARD NUMBER BACB30VF ⁸	TL (NOM) ⁵	TD ⁶		U MAX	Ø W GAGE +.0002 -.0000 ³	TOL Y	TOL Z	RECESS NUMBER PER NAS7100 ¹⁶	RECESS TORQUE IN-LB MIN	TENSILE STRENGTH LBS MIN
		MAX	MIN							
6	.454	.3680	.3640	.042	.4852	.0060	.0025	4	230	6,480
7	.528	.4310	.4260	.050	.5696				300	9,000
8	.528	.4930	.4880		.6498		.0020		440	12,150

NOTES

- ¹ TO DETERMINE THE NOMINAL GRIP LENGTH, MULTIPLY THE GRIP LENGTH NUMBER (SECOND DASH NUMBER) BY .0625. THE GRIP LENGTH IS MEASURED FROM THE TOP OF THE HEAD TO THE END OF THE FULL CYLINDRICAL PORTION OF THE SHANK.
- ² TO DETERMINE THE GRIP LENGTH NUMBER, DIVIDE THE TOTAL THICKNESS OF PARTS BEING JOINED BY .0625. ROUND OFF DECIMALS TO NEXT LARGER WHOLE NUMBER.
- ³ INSPECTION OF HEAD PROTRUSION "H" AT GAGE DIAMETER "W" SHALL CONFORM TO BOEING DOCUMENT [D-11805](#).
- ⁴ TRUE POSITION WITHIN .016 DIAMETER WITH RESPECT TO DATUM "B" FOR SIZES 4 (.250 DIAMETER) AND SMALLER.
- ⁵ TOLERANCE SHALL BE APPLIED TO A NOMINAL LENGTH DETERMINED BY ADDING THE NOMINAL GRIP LENGTH (SEE ¹) TO THE NOMINAL THREAD LENGTH, "TL" FROM [TABLE I](#) ("TL" IS A NOMINAL DIMENSION AND IS NOT TO BE INSPECTED).
- ⁶ THREAD PER [AS8879](#), CLASS UNJF-3A EXCEPT CLASS UNJC-3A FOR SIZE 08; REDUCED MAJOR DIAMETER PER "TD". INCOMPLETE THREADS AT CHAMFER AND GRIP SHALL CONFORM TO [BPS-F-69](#).
- ⁷ END SHALL BE FLAT AND CHAMFERED PER [BPS-F-69](#) EXCEPT CHAMFER PER "U" IN [TABLE I](#).
- ⁸ SEE CODING UNDER USAGE AND APPLICATION SECTION FOR COMPLETE BOEING PART NUMBER.
- ⁹ ON OVERSIZE FOR BACB30VF08() USE BACB30VF3() FOR FIRST OVERSIZE AND BACB30VF3()X FOR SECOND OVERSIZE.
- ¹⁰ "A", "A' ", AND "B" ARE FOR ENGINEERING REFERENCE ONLY AND ARE NOT TO BE USED FOR INSPECTION. THESE ARE CALCULATED LIMITS RESULTING FROM TOLERANCES ON "D", "E", "H", "W" AND HEAD ANGLE.

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BACB30VF

SH 6

BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)

BACB30VF

SH 6

BOEING PART STANDARD

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FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

NOTES (CONTINUED)

- 11** RUNOUT OF "D" DIAMETER TO THREAD PITCH DIAMETER WITHIN "Y" FIM WHEN HELD ON THE PITCH DIAMETER OF THE COMPLETE THREADS NEAREST THE SHANK AND CHECKED ON THE SHANK WITHIN ONE DIAMETER OF THE THREAD RUNOUT.
- 12** SEE INACTIVATION APPLICABILITY BLOCK.
- 13** DIMENSIONS APPLY TO "B" CODE PARTS ONLY.
- 14** "B" CODE TO BE USED WITH "X" CODE PARTS, ONLY.
- 15** RESTRICTED USAGE: REPAIR ONLY.
- 16** PARTS MANUFACTURED PRIOR TO 16-JAN-2013 MAY CONTAIN A RECESS CONFORMING TO ANY REVISION OF [NAS7100](#). PARTS MANUFACTURED AFTER 16-JAN-2013 SHALL BE MANUFACTURED USING THE [NAS7100](#) REVISION IN EFFECT ON THE DATE OF MANUFACTURE.
- 17** HUCK LAKEWOOD (FORMERLY DEUTSCH FASTENER CORP) IS NO LONGER IN OPERATION. SIZES 08 THRU 8, MANUFACTURED AT THAT FACILITY (CAGE CODE 97928) UNDER EITHER NAME PRIOR TO OCTOBER 1, 1999 MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- 18** PARTS MARKED WITH BLACK DYE OR PAINT ON THREAD END, MANUFACTURED BEFORE JANUARY 01, 2004, MAY BE PROCURED AND USED UNTIL INVENTORY IS DEPLETED.
- 19** FOR "G" AND "H" CODED PARTS ONLY, HEAD SURFACES INDICATED ARE PAINTED. SEE FINISH SECTION. NO OVERSPRAY PERMITTED. ALL DIMENSIONS APPLY PRIOR TO THE APPLICATION OF PAINT UNLESS OTHERWISE SPECIFIED.
- 20** ALCOA FASTENING SYSTEMS (TEMPLE) (FORMERLY FAIRCHILD - TEMPLE) IS NO LONGER IN OPERATION. PARTS MANUFACTURED AT THAT FACILITY (CAGE CODE 06950) UNDER EITHER NAME PRIOR TO SEPTEMBER 1, 2006 MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- 21** FAIRCHILD FASTENERS (UNRUH) IS NOW ALCOA FASTENING SYSTEMS - UNRUH. THIS WAS A NAME CHANGE ONLY. PARTS MANUFACTURED UNDER THE FAIRCHILD NAME MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- 22** HUCK INTERNATIONAL INC (CARSON) IS NOW ALCOA FASTENING SYSTEMS - CARSON. THIS WAS A NAME CHANGE ONLY. PARTS MANUFACTURED UNDER THE HUCK NAME MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- 23** SPS TECHNOLOGIES (KEARNS) IS NO LONGER IN OPERATION. PARTS MANUFACTURED AT THAT FACILITY (CAGE CODE 0PTK6) PRIOR TO JUNE 30, 2002 MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.

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BACB30VF**SH 7**

**BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)**

BACB30VF**SH 7****BOEING PART STANDARD**

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NOTES (CONTINUED)

- [24] FOR "E" CODED PARTS ONLY, HEAD AND THREAD SURFACES INDICATED SHALL BE ALUMINUM PIGMENTED COATED (SEE FINISH SECTION). OVERSPRAY OF UP TO 0.005 INCH IS ALLOWED ON THE CONICAL SURFACE OF THE HEAD. INCOMPLETE COVERAGE OF THE ALUMINUM PIGMENTED COATING IS ALLOWED ON THE RUNOUT THREADS STARTING FROM THE GRIP SECTION OF THE BOLT SHANK, EXTENDING INTO THE THREADS A MAXIMUM OF 0.030 INCH. NO OVERSPRAY IS ALLOWED ON THE BOLT SHANK.
- [25] "K" CODED PARTS WITH [BMS10-85](#), TYPE I, CLASS B COATING MANUFACTURED PRIOR TO SEPTEMBER 1, 2004 MAY BY PROCURED AND USED UNTIL DEPLETED.

PROCUREMENT SPECIFICATION

[BPS-F-69](#), CLASS 95ST3 OR 95ST4, EXCEPT AS NOTED.

ACCEPTANCE TESTS

THE FOLLOWING TESTS SHALL BE PERFORMED IN ADDITION TO TESTING REQUIRED BY [BPS-F-69](#):

BOLT RECESS SHALL BE TORQUE TESTED IN BOTH INSTALLATION AND REMOVAL DIRECTIONS. USE THE APPROPRIATE SIZE DRIVER IN ACCORDANCE WITH NAS8500 THRU 8506 WITH AXIAL END LOAD NOT EXCEEDING 15 LBS. BOLTS ARE REJECTABLE IF MINIMUM TORQUE VALUES LISTED IN [TABLE I](#) CAUSE FRACTURE OF BOLT OR IN RAISED METAL AT EDGE OF RECESS EXCEEDING .005 ABOVE SURROUNDING AREA. SAMPLING FOR LOT INSPECTION OF RECESS TORQUE SHALL BE IN ACCORDANCE WITH [TABLE II](#) OF THE PROCUREMENT SPECIFICATION.

TENSILE STRENGTH SHALL CONFORM TO THE REQUIREMENTS OF [TABLE I](#) AND [TABLE II](#). SAMPLING FOR LOT INSPECTION OF TENSILE STRENGTH SHALL BE IN ACCORDANCE WITH THE PROCUREMENT SPECIFICATION.

MATERIAL

6Al-4V TITANIUM ALLOY PER [AMS4928](#) OR [AMS4967](#).

FINISH

ALUMINUM PIGMENTED COATING PER [BMS10-85](#), TYPE I, CLASS A. [25]

PHOSPHATE-FLUORIDE TREAT PER [BAC5861](#).

ALUMINUM PIGMENTED COATING PER [BMS10-85](#), TYPE I, CLASS A. [25]
SURFACES INDICATED BY [19] SHALL BE PAINTED AS FOLLOWS:

"G" CODE:

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BACB30VF

SH 8

**BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)**

BACB30VF

SH 8

BOEING PART STANDARD

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FOR STATUS OF INACTIVATION
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FINISH (CONTINUED)

- 1 SOLVENT CLEAN IN ACCORDANCE WITH [BAC5750](#).
- 2 APPLY ONE COAT OF [BMS10-79](#), TYPE III PRIMER IN ACCORDANCE WITH [BAC5882](#).
- 3 APPLY [BMS10-60](#), TYPE II, BOEING COLOR 707 GREY GLOSS ENAMEL IN ACCORDANCE WITH [BAC5845](#).

"H" CODE:

- 1 SOLVENT CLEAN IN ACCORDANCE WITH [BAC5750](#).
- 2 APPLY ONE COAT OF [BMS10-11](#), TYPE I PRIMER IN ACCORDANCE WITH [BAC5736](#).
- 3 APPLY [BMS10-83](#), TYPE V, BOEING COLOR 8925 BEIGE FLAT ENAMEL IN ACCORDANCE WITH [BAC5755](#).

SURFACE TEXTURE

PER [ASME B46.1](#), HEAD TO SHANK FILLET, "D" DIAMETER, CONICAL SURFACE OF HEAD, THREAD FLANKS AND THREAD ROOTS, 32 MICROINCHES Ra; OTHER SURFACES, 125 MICROINCHES Ra.

LUBRICATION

CETYL ALCOHOL PER [AS87132](#), TYPE I, GRADE OPTIONAL, OR TYPE III, GRADE B.

MARKING [18](#)

FOR .2500 NOMINAL DIAMETER AND SMALLER, MARK WITH "VF", FIRST DASH NUMBER, AND MANUFACTURER'S SYMBOL/INSIGNIA PER [MIL-HDBK-57](#) OR REGISTERED WITH THE U.S. PATENT AND TRADEMARK OFFICE (PTO) OF THE U.S. DEPARTMENT OF COMMERCE. FOR .3125 NOMINAL DIAMETER AND LARGER, MARK WITH "B30VF", FIRST DASH NUMBER, AND MANUFACTURER'S SYMBOL/INSIGNIA. FOR OVERSIZE FASTENERS, MARK "X" OR "Y" AS SPECIFIED. MARKING SHALL BE DEPRESSED .010 MAXIMUM, ARRANGEMENT OPTIONAL.

PHOSPHATE-FLUORIDE TREATED BOLTS SHALL BE IDENTIFIED WITH RED DYE OR PAINT ON THE THREAD END. "E" CODED PARTS SHALL BE IDENTIFIED WITH WHITE PAINT ON THE THREAD END. MAXIMUM COVERAGE MAY INCLUDE THE CHAMFER DIMENSION PLUS ONE PITCH.

PROCUREMENT [17](#) [20](#) [23](#)

AIR INDUSTRIES CORPORATION (CAGE CODE 06725)
(FOR SIZES 08 THRU 4 ONLY)

ALCOA FASTENING SYSTEMS - CARSON (CAGE CODE 17446) [22](#)

DATE 02-JUN-1982 REV (Y) 04-NOV-2013

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BACB30VF**SH 9**

**BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)**

BACB30VF**SH 9****BOEING PART STANDARD**

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PROCUREMENT [17](#) [20](#) [23](#) (CONTINUED)ALCOA FASTENING SYSTEMS - UNRUH (CAGE CODE 1RC86) [21](#)

HI-SHEAR CORPORATION (CAGE CODE 73197)

PB FASTENERS, DIVISION OF PAUL R BRILES, INC (CAGE CODE 27624)

THE MANUFACTURERS LISTED IN [BPS-F-69SUP](#) AND THEIR AUTHORIZED DISTRIBUTORS ARE THE ONLY APPROVED SOURCES FOR THE ABOVE QUALIFIED PRODUCTS. SEE [BPS-F-69SUP](#) FOR PLANT ADDRESSES. NO CHANGES IN PRODUCT DESIGN, BASIC METHODS OF MANUFACTURE, PLANT SITE OR QUALITY LEVEL SHALL BE MADE WITHOUT PRIOR NOTIFICATION AND PRIOR APPROVAL IN WRITING FROM THE BOEING COMPANY. MANUFACTURERS OF COMPETITIVE PRODUCTS MAY APPLY TO A SUPPLIER MANAGEMENT DEPARTMENT OF THE BOEING COMPANY FOR QUALIFICATION. IF A MANUFACTURER IS SHOWN ON THIS STANDARD, BUT IS NOT LISTED IN THE SUPPLEMENT, CONTACT THE DIVISIONAL ENGINEERING STANDARDS FOCAL POINT OR ENGINEERING STANDARDS FOR VERIFICATION.

USAGE AND APPLICATION INFORMATION

THESE BOLTS ARE NOT RECOMMENDED FOR USE IN APPLICATIONS WHICH REQUIRE FREQUENT INSTALLATION AND REMOVAL OR WHERE HIGH JOINT CLAMP-UP IS REQUIRED.

OVERSIZE BOLTS, CODES "X" AND "Y", ARE TO BE USED AS REPLACEMENTS FOR STANDARD SIZE BACB30VF BOLTS WITH THE SAME FINISH WHEN THE HOLE FOR THE STANDARD SIZE BOLT IS OVERSIZE OR MUST BE MADE OVERSIZE TO CORRECT DAMAGE OR MISALIGNMENT. [15](#)

THE COUNTERSINK MUST BE ENLARGED FOR BACB30VF() () () Y AND BACB30VF() B () () X BOLTS BECAUSE THESE BOLTS HAVE LARGER HEAD DIAMETERS THAN BACB30VF() B () () X AND BACB30VF() () () X BOLTS OR BACB30VF() () () X AND NOMINAL SIZE BOLTS, RESPECTIVELY.

THE TENSILE VALUES LISTED IN [TABLE I](#) ARE FOR LOT ACCEPTANCE INSPECTION TESTING PURPOSES ONLY. DESIGNERS SHALL USE THE BOEING DESIGN MANUAL, [BDM-4282](#) FOR ALLOWABLES.

SEE [BACN10JC](#) FOR MATING NUT.

INSTALL PER [BAC5009](#) OR [BAC5063](#), AS APPLICABLE.

MAXIMUM SERVICE TEMPERATURE

250 F ALUMINUM PIGMENTED COATED BOLTS, PAINTED HEAD.

350 F ALUMINUM PIGMENTED COATED BOLTS, NO PAINT.

450 F PHOSPHATE-FLUORIDE TREATED BOLTS.

DATE 02-JUN-1982 REV (Y) 04-NOV-2013

CAGE CODE 81205

BACB30VF**SH 10**

**BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)**

BACB30VF**SH 10****BOEING PART STANDARD**

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

MAXIMUM SERVICE TEMPERATURE (CONTINUED)**CAUTION**

TEST DATA INDICATES THAT PHOSPHATE FLUORIDE COATED FASTENERS INSTALLED IN CARBON-EPOXY STRUCTURES CAN BE POTENTIALLY HARMFUL TO THE STRUCTURE IF EXPOSED TO A SALT ENVIRONMENT.

SEE [BACD2074](#) FOR FASTENER CODES.

CODING

FIRST DASH NUMBER DESIGNATES NOMINAL THREAD SIZE PER [TABLE I](#) OR [TABLE II](#).

LETTER "B" FOLLOWING THE FIRST DASH NUMBER DESIGNATES AN ENLARGED DIAMETER HEAD. CODE "B" IS ONLY USED FOR THE FIRST OVERSIZE (CODE "X") PARTS AND IS IN ADDITION TO, AND PRIOR TO, THE "E", "K" OR "R" FINISH CODES.

LETTER "E" FOLLOWING FIRST DASH NUMBER DESIGNATES A BARE BOLT WITH ALUMINUM PIGMENTED COATING ON THE TOP OF THE HEAD AND THE THREADS, AND NO LUBRICATION.

LETTER "K" FOLLOWING THE FIRST DASH NUMBER DESIGNATES ALUMINUM PIGMENTED COATING AND NO LUBRICATION. (CLASS 95ST3)

LETTER "R" FOLLOWING THE FIRST DASH NUMBER DESIGNATES PHOSPHATE-FLUORIDE FINISH WITH CETYL ALCOHOL LUBRICATION. (CLASS 95ST4)

SECOND DASH NUMBER DESIGNATES NOMINAL GRIP LENGTH IN .0625 INCREMENTS. [2](#)

LETTER "G" FOLLOWING THE SECOND DASH NUMBER DESIGNATES BOEING COLOR 707 GRAY ENAMEL PAINTED HEAD. [19](#)
(USE WITH ALUMINUM PIGMENT COATED BOLTS ONLY)

LETTER "H" FOLLOWING THE SECOND DASH NUMBER DESIGNATES BOEING COLOR 8925 BEIGE FLAT ENAMEL PAINTED HEAD. (USE WITH ALUMINUM PIGMENT COATED BOLTS ONLY). [19](#)

LETTER "X" AT THE END OF THE PART NUMBER DESIGNATES .0156 OVERSIZE SHANK. [15](#)

LETTER "Y" AT THE END OF THE PART NUMBER DESIGNATES .0312 OVERSIZE SHANK. [15](#)

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BACB30VF**SH 11**

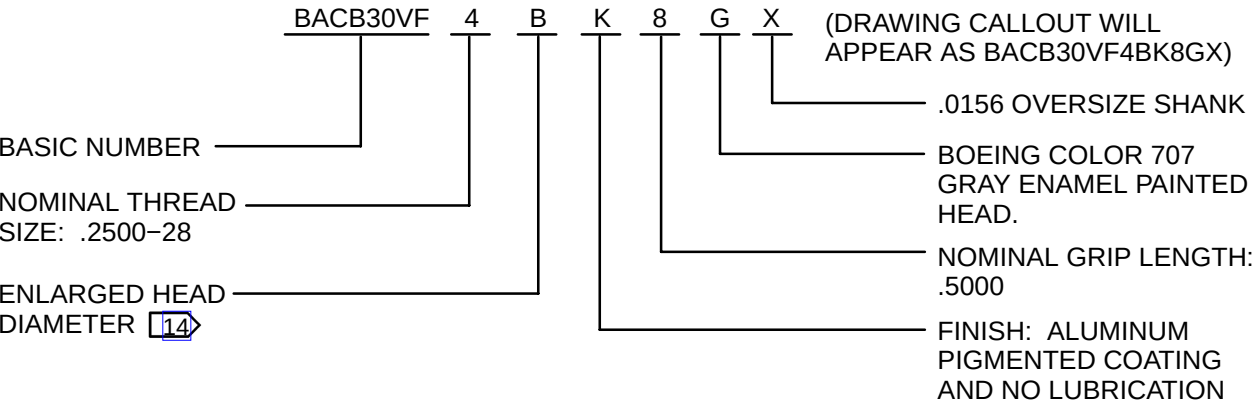
**BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)**

BACB30VF**SH 11****BOEING PART STANDARD**

***** PSDS GENERATED *****

FOR STATUS OF INACTIVATION
SEE APPLICABILITY BLOCK

EXAMPLE OF PART NUMBER



SEE [D-590-PREFACE](#) (INDEX) FOR INACTIVATION DEFINITIONS. SEE [D-590-SUPERSESSSION-LIST](#) FOR SUPERSESSSION CLASS DEFINITIONS AND SUPERSESSSION LIST.

INACTIVATION APPLICABILITY

- BCA, BDS - BACB30VF()K()(NO CODE, G)X ARE INACTIVE FOR DESIGN AND PROCUREMENT.
BACB30VF()BK()(NO CODE, G)X ARE CLASS II SUPERSESSSIONS, RESPECTIVELY.
- BACB30VF()R()X IS INACTIVE FOR DESIGN AND PROCUREMENT.
BACB30VF()BR()X IS A CLASS II SUPERSESSSION.
- BCA - BACB30VF()R()(NOMINAL, Y) ARE INACTIVE FOR NEW DESIGN.

DATE 02-JUN-1982 REV (Y) 04-NOV-2013

CAGE CODE 81205

BACB30VF
SH 12

**BOLT, 100 DEG REDUCED HEAD, CROSS
RECESS, 95 KSI SHEAR, SHORT THREAD,
6AL-4V TITANIUM (NOMINAL AND OVERSIZE)**

BACB30VF
SH 12

BOEING PART STANDARD